Solar Al appendix

Sun · Al · downloadable appendix

Document	Solar Al appendix
Subject	Solar photospheric abundance appendix · Al
Author	Ryan Schmitt
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Solar aluminium: 6.541 ± 0.037

Asplund, Amarsi & Grevesse 2021: 6.43 ± 0.03 · this measurement +0.111 dex

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (Nordlander & Lind 2017: the Al I lines of their Table A.1), 6 lines measured on one spectrum, each part weighted by its uncertainty.

Al I: 6.516 ± 0.037 · Kitt Peak (1984, molecfit) · VIS · Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber)

Al I: 6.521 ± 0.051 · Kitt Peak (1984, molecfit) · red-optical · Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber)

Al I: 6.591 ± 0.044 · Kitt Peak (1984, molecfit) · NIR · Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber)

Highlighted Products

VIS

6.488 ± 0.037

Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Reference Grade

red-optical

6.489 ± 0.034

Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · IAG FTS · Reference Grade

NIR

6.591 ± 0.044

Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Reference Grade

Line grades

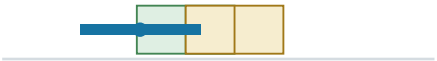
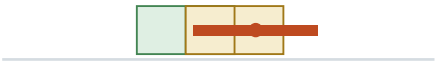
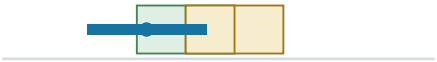
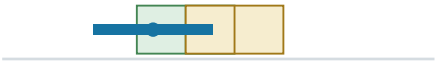
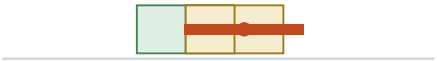
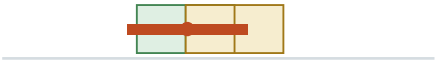
Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	46 products -- Al I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	40 products -- Al I · Codex-graded lines

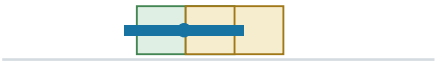
Grade	What it means	This element
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

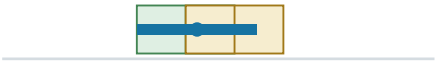
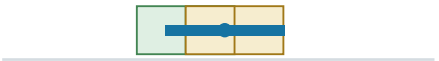
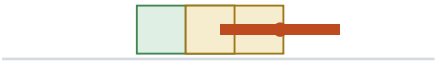
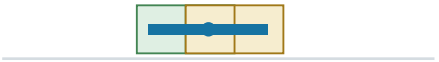
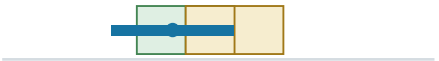
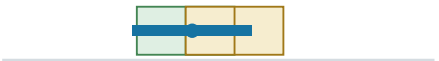
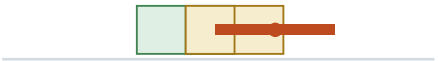
VIS — HARPS
A(Al) dex; axis 6.32–6.58

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.413 ±0.038
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.391 ±0.037
Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.402 ±0.037
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.417 ±0.038
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.473 ±0.038
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.446 ±0.038
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.406 ±0.037
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.384 ±0.037
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.396 ±0.037
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.410 ±0.037
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.466 ±0.037
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.431 ±0.037

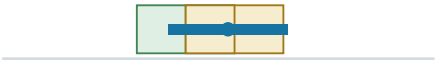
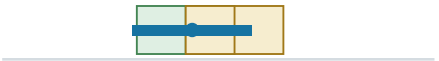
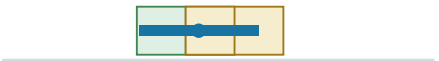
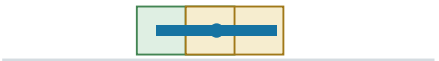
VIS — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.32–6.58

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.451 ±0.037
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.429 ±0.037

VIS — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.32–6.58

Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.437 ±0.037
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.454 ±0.037
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.511 ±0.037
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.488 ±0.037
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.444 ±0.037
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.422 ±0.038
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.434 ±0.037
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.447 ±0.037
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.504 ±0.037
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.485 ±0.037

VIS — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.32–6.58

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.456 ±0.037
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.434 ±0.037
Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.438 ±0.037
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.460 ±0.037
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.516 ±0.037
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.490 ±0.037
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.449 ±0.037

VIS — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.32–6.58

Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.427 ±0.038
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines	6.429 ±0.037
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines	6.453 ±0.037
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.508 ±0.037
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.480 ±0.037

VIS axis: 6.32 · 6.38 · 6.45 · 6.52 · 6.58 A(Al) dex

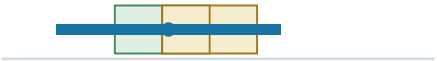
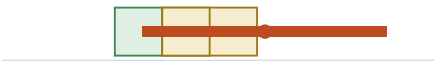
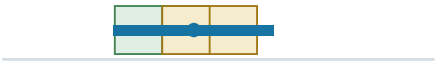
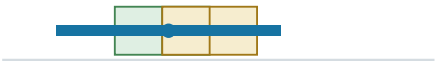
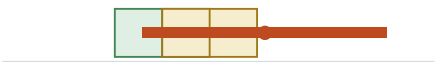
red-optical — IAG FTS
A(Al) dex; axis 6.33–6.60

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.437 ±0.035
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.419 ±0.037
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines	6.438 ±0.036
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.489 ±0.034
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines	6.437 ±0.035
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.419 ±0.037
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines	6.438 ±0.036
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.489 ±0.034

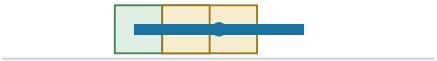
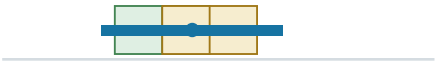
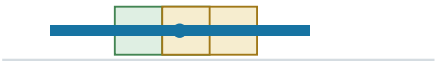
red-optical — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.33–6.60

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.467 ±0.047
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.450 ±0.051

red-optical — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.33–6.60

Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.434 ±0.071
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.470 ±0.049
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.522 ±0.045
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.495 ±0.077
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.467 ±0.047
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.450 ±0.051
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.434 ±0.071
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.470 ±0.049
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.522 ±0.045
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.495 ±0.077

red-optical — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.33–6.60

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.466 ±0.053
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.449 ±0.057
Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines		6.441 ±0.082
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines		6.469 ±0.055
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.521 ±0.051
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.488 ±0.080
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.466 ±0.053

red-optical — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.33–6.60

Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.449 ±0.057
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines	6.441 ±0.082
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines	6.469 ±0.055
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.521 ±0.051
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED	6.488 ±0.080

red-optical axis: 6.33 · 6.40 · 6.47 · 6.53 · 6.60 A(Al) dex

NIR — CRIRES+ Y (full arm)
A(Al) dex; axis 6.13–7.51

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line	6.837 ±0.088
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line	6.822 ±0.088
Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 1 line	6.798 ±0.638
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 1 line	6.838 ±0.088
Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 1 line · EXPERIMENTAL-NOT-ADOPTED	6.884 ±0.088
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 1 line · EXPERIMENTAL-NOT-ADOPTED	6.845 ±0.636

NIR — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.13–7.51

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.537 ±0.040
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.520 ±0.039
Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 2 lines	6.543 ±0.063
Al I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 2 lines	6.540 ±0.040

NIR — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.13–7.51

Al I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.591 ±0.044
Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.580 ±0.073
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.537 ±0.040
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.520 ±0.039
Al I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		6.543 ±0.063
Al I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		6.540 ±0.040
Al I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.591 ±0.044
Al I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		6.580 ±0.073

NIR axis: 6.13 · 6.48 · 6.82 · 7.17 · 7.51 A(Al) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every aluminium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the aluminium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	VIS
What was measured	Al I · Codex-graded lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Al)	6.466–6.508 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.06, model atmosphere 0.02

Band	VIS
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Al)	6.446–6.490 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.06, model atmosphere 0.02

Band	red-optical
What was measured	Al I · Codex-graded lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Al)	6.489–6.522 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.05, model atmosphere 0.02

Band	red-optical
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Al)	6.489–6.522 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.05, model atmosphere 0.02

Band	NIR
What was measured	Al I · Codex-graded lines
Solar spectra	Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	2
Result A(Al)	6.591 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.06, fit / line-to-line scatter 0.03

Band	NIR
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	1–2
Result A(Al)	6.591–6.884 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	line strengths (gf) 0.06, fit / line-to-line scatter 0.03

Solar aluminium context

Aluminium is traced by Al I lines in the optical and near-infrared. The reference lines are the six Asplund et al. 2021 used (Nordlander & Lind 2017: 6696, 6698, 7835, 8912, 10768 and 10872 Å) with the oscillator strengths and uncertainties those authors adopted. Al I departs from LTE, so each spectrum also carries a non-LTE-corrected product; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Al appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 078cf7dedfa961452c8bd9af43c4bf83c5463fdf
Feed Al.json v1.159 · updated 2026-11-08T00:07:39Z
Generated 2026-11-09T08:19:44-07:00 (source commit timestamp)
Feed SHA-256 2501309925b7168558c4b6b497992edd4a788fe98a67266df27d9542afbad1c7
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Al comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set AL_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Al I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Al_Amarsi2020_PySME.csv). Sources: [5].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: synth-mean3D-LTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

Model and grid sources. Feed treatment: synth-mean3D-NLTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

Product-to-source index

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-3dced287aa93

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-6a4b66082042

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Al-products-0656fc282fb7

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NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Al-products-4fbec410b2fc

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red-optical · IAG FTS · Al I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [3], [4]
Al-quarantine-894223314462

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